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RESEARCH ARTICLE

To plant or not to plant? Nutrient uptake, growth, and biochemistry of eucheumatoids under low salinity as influenced by nutrient-rich freshwater outflow

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Abstract

Seaweeds can be negatively affected by low salinity, but the negative impacts may be ameliorated by the presence of high nutrient levels from the freshwater inputs. We tested this hypothesis on the eucheumatoids *Kappaphycus alvarezii*, *K. striatus*, and *Eucheuma denticulatum* by determining their nutrient uptake, growth, and biochemical responses to experimental treatments of lowering salinity with a consequent increase in nutrient concentrations. Ambient seawater with varying salinity and nutrient concentrations was sourced offshore (O) and nearshore (N) and diluted with nutrient-replete natural spring water to obtain six (6) treatments: O-LN-34, O-MN-30, O-HN-27, N-MN-30, N-HN-27, and N-HN-24 (corresponding to water source-relative nutrient concentration- salinity, where LN = low nutrient; MN = medium nutrient; HN = high nutrient). Nutrient uptake varied among inorganic nutrient species and different eucheumatoid species. All eucheumatoids showed uptake preference for NH_4^+ regardless of the treatment while NO_3^- uptake was generally higher in treatments with higher NO_3^- concentration. Growth rates were species-specific but generally higher at lower salinities with higher nutrient concentrations. Total soluble carbohydrate, total protein, and total polyphenol content did not significantly vary among treatments, except for *E. denticulatum*. These results provide insights into the potential of farming eucheumatoids nearshore with fluctuating salinity provided that the freshwater outflow is nutrient-rich to ameliorate the negative effects of low salt concentration.

KEYWORDS
Eucheuma denticulatum, *Kappaphycus alvarezii*, *Kappaphycus striatus*, nutrient physiology, seaweed farming, stress tolerance

INTRODUCTION

Eucheumatoids (Rhodophyta), also known as carrageenophytes, are farmed for carrageenan—a polysaccharide with a multitude of applications in many industries such as pharmaceutical, food, and cosmetic, among others (Rupert et al., 2022). Their farming is based primarily on the cultivation of the two genera,

Abbreviations: AlgaE Lab, Algal Ecophysiology Laboratory; ANOVA, analysis of variance; BARC, Bolinao-Anda Reef Complex; BML, Bolinao Marine Laboratory; DBL, diffusion boundary layer; DIC, dissolved inorganic carbon; DOM, dissolved organic matter; DON, dissolved organic nitrogen; HN, high nutrient; LN, low nutrient; NRFW, nutrient-replete freshwater; OLBN, outdoor land-based nursery; SCL&GB, Seaweed Culture Laboratory and Gene Bank; SGD, submarine groundwater discharge; SW, seawater; TIN, total inorganic nitrogen; TPC, total polyphenol content; UPMSI, University of the Philippines—The Marine Science Institute.

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Kappaphycus and *Eucheuma* (Valderrama et al., 2013). In many countries, *Kappaphycus* spp. and *Eucheuma denticulatum* are intensively farmed, accounting for ~35%–45% of the reported total global seaweed aquaculture production (Buschmann et al., 2017; Food and Agricultural Organization of the United Nations, 2016; Zhang et al., 2022).

When establishing a seaweed farm, environmental conditions—for example, seawater salinity, turbidity, and temperature, among others—of the cultivation site are key considerations. Conventional eucheumatoid seaweed farming discourages planting in sites influenced by freshwater outflows due to the negative impacts of low salinity (Jaunich, 1988; Trono Jr., 1992). Therefore, most of the traditional farming sites of *Kappaphycus alvarezii, K. striatus*, and *Eucheuma denticulatum* in the Philippines are situated in areas with seawater salinity greater than 30 (Hurtado et al., 2001; Luhan & Sollesta, 2010; Melendres & Largo, 2021; Tahiluddin et al., 2022). The same salinity requirement has been adopted across other tropical and subtropical countries farming eucheumatoids (e.g., Indonesia, Brazil, and Tanzania: Hayashi et al., 2007; Msuya & Porter, 2014; Simatupang et al., 2021). This traditional practice may constrain the expansion of seaweed farming in the shallow subtidal and lower intertidal that are easily accessible to women involved in seaweed farming but periodically exposed to environmental stressors like high light and temperature, desiccation stress, and fluctuating salinity due to rainfall and freshwater outflow during low tides.

Salinity is among the abiotic factors that influence seaweed growth, survival, and distribution range in aquatic environments. It affects vital physiological processes such as photosynthesis, respiration, and growth, among others (Hurd et al., 2014; Karsten, 2012, 2024). Generally, changes in salinity beyond the tolerable range of macroalgae may result in damage to membranes, organelles, or enzymes (Hurd et al., 2014). However, macroalgae exposed to low salinity may synthesize organic solutes that aid in osmoregulation (Hurd et al., 2014; Karsten, 2012, 2024). For rhodophytes, floridoside is synthesized by most members of this taxa (Gambichler et al., 2021; Karsten, 2012, 2024), although some red algae—for example, *Bostrychia calliptera*, *B. arbuscula*, and *B. montagnei*—synthesize dulcitol and sorbitol as organic osmolytes (Borburema et al., 2022; Gambichler et al., 2021). Additionally, inorganic nutrient ions, particularly NO_3^-, PO_4^{3-}, and SO_4^{2-}, are also taken up by macroalgae as inorganic osmolytes that aid in osmotic regulation and eventually contribute to their nutrient requirements (Cohen & Fong, 2004; Karsten, 2024). Consequently, the independent negative effects of low salinity can be alleviated under nutrient-replete conditions.

Surprisingly, most of the available salinity response studies on eucheumatoids prepare lower salinity medium by diluting ambient seawater with distilled or deionized freshwater, that is, stripped of important minerals, trace elements, and macronutrients (e.g., Hayashi et al., 2011; Reis et al., 2011) compared with natural freshwater outflow (this study), submarine groundwater discharge (commonly termed as SGD), and rainwater. Other studies have utilized freezing and gradual melting of seawater, resulting in seawater with different salinities and mixing to obtain different salinity treatments (e.g., Araújo et al., 2014; however, the inorganic nutrient concentrations resulting from this process were, unfortunately, not reported). Without considering the chemistry of the resulting lower salinity medium, that is, primarily with consequent lower macronutrient (N, P) concentration compared to the ambient seawater used as control, results of those salinity response studies are ambiguous, as the negative hyposalinity effect may have been exacerbated by the lower concentrations of minerals and macronutrients.

Incidentally, recent field observations and anecdotal accounts from the seaweed farmers in northern Bohol, Philippines, have provided evidence that seaweed farming in areas near mangrove forests with freshwater outflow, presumptive higher inorganic and organic nutrients, and high turbidity elicits no negative effects. In fact, farmers observed higher growth rates of eucheumatoids in these areas. Nutrient levels in coastal marine environments are enriched by various allochthonous sources such as freshwater outflows from rivers or natural springs and SGDs. For instance, rivers transport inorganic nutrients and terrigenous organic matter in coastal marine habitats, the latter then becoming an important source of inorganic nutrients after undergoing microbial degradation (Ferrera et al., 2016; Guo & Macdonald, 2006; Kamjunke et al., 2023; Schlünz & Schneider, 2000). Additionally, mangrove forests are also an important source of inorganic nutrients, particularly ammonium, to nearby coastal environments (Taillardat et al., 2019; Wang et al., 2019). Although freshwater outflows into the coastal marine environment have led to hyposaline conditions, they have, however, increased the amount of inorganic nutrients in the water. These alterations in the seawater chemistry will consequently affect macroalgal physiology as well as other metabolic processes vital for macroalgal growth and survival. Whether the eucheumatoids will positively respond to lower salinity with a consequent increase in nutrient concentration warrants rigorous and comprehensive investigation. A mechanistic manipulation of different salinities with different macronutrient concentrations was used to test our hypothesis that natural nutrient-rich freshwater outflows reducing coastal water salinity will elicit minimal to no significant negative effects on the growth and biochemistry of three commercially important eucheumatoid species.

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MATERIALS AND METHODS

Seawater and freshwater collection sites

Seawater collection sites are shown in Figure 1. Medium-nutrient (MN) seawater with an ambient salinity of 30 was sourced nearshore along the Guiguiwanen Channel, Bolinao, Pangasinan, Philippines. The eutrophic water here is the consequence of intensive fish farming for decades (San Diego-McGlone et al., 2008; Ferrera et al., 2016). Conversely, low-nutrient (LN) seawater with an ambient salinity of 34 was collected offshore in the Malilnep Channel north of the Bolinao-Anda Reef Complex (BARC). This site is ~6.36 km away from the Guiguiwanen Channel. The nutrient-replete freshwater (NRFW) containing high inorganic nutrient levels was collected from a natural spring, locally called *subor*, near the beach area.

Seawater in each site was collected at 1 m depth using a Niskin water sampler from the two channels in Bolinao, Pangasinan. Collected seawater was transferred to clean 20-L black containers mounted with a 200-μm filter bag and was brought to the laboratory for culture media preparation. Freshwater was collected directly from the spring source using a water dipper and then carefully transferred to a clean 20-L container mounted with a 200-μm filter.

Preparation of seawater culture media

To maintain the natural carbonate chemistry of the water, all collected water samples were not autoclaved. To obtain lower salinity values, the collected NRFW was used to dilute the seawater from Malilnep Channel and Guiguiwanen Channel. The seawater (SW) from Malilnep with a salinity of 34 was diluted with NRFW using 9:1 and 4:1 SW:NRFW ratios to obtain culture media with 30 and 27 salinities, respectively. Similarly, the SW from Guiguiwanen with a salinity of 30 was diluted to 13:2 and 11:4 SW:NRFW ratios, resulting in culture media with 27 and 24 salinities, respectively. Salinity was measured using a calibrated ATAGO handheld refractometer. The above salinities correspond to 10%–11% and 20% salinity reductions from the original ambient salinities. Following the dilution process,

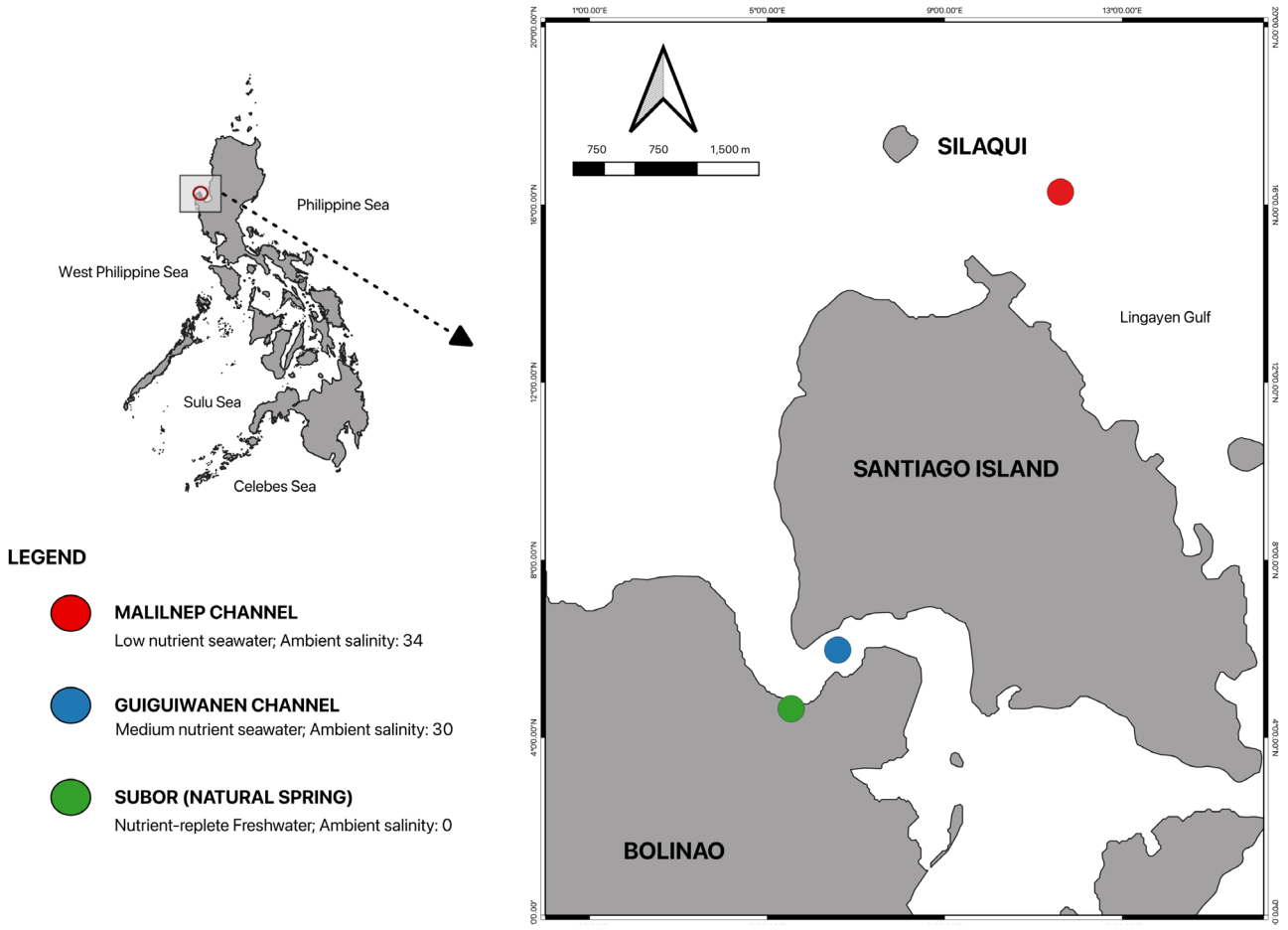


FIGURE 1 Map of the seawater and freshwater collection sites in Bolinao, Pangasinan. The map was generated using QGIS 3.36. The shapefile data of the Philippine map was downloaded from https://data.humdata.org/dataset/cod-ab-phl. Site coordinates were obtained from Google Earth.

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TABLE 1 Nutrient concentration (μM) of the different salinity-nutrient treatments (mean ± *SE*).

Seawater source	% salinity reduction	Salinity	Nutrient concentration (μM)					Salinity-nutrient treatments
			PO_4^{3-}	NH_4^+	NO_2^-	NO_3^-	Total N	
Offshore Low Nutrient								
O-LN	0%	34	0.60 ± 0.04	1.45 ± 0.24	0.18 ± 0.04	0.72 ± 0.11	2.35	O-LN-34
O-LN	10%	30	0.56 ± 0.09	1.25 ± 0.10	0.16 ± 0.05	3.89 ± 1.54	5.30	O-MN-30
O-LN	20%	27	0.58 ± 0.10	2.30 ± 0.09	0.16 ± 0.03	9.41 ± 0.44	11.88	O-HN-27
Nearshore High Nutrient								
N-HN	0%	30	1.89 ± 0.09	5.50 ± 0.10	0.33 ± 0.04	0.47 ± 0.05	6.30	N-MN-30
N-HN	11%	27	1.76 ± 0.08	5.41 ± 0.09	0.26 ± 0.05	8.26 ± 0.70	13.93	N-HN-27
N-HN	20%	24	1.61 ± 0.24	5.44 ± 0.14	0.34 ± 0.05	8.76 ± 1.94	14.56	N-HN-24
Nutrient-replete freshwater (NRFW)		0	0.80 ± 0.06	2.56 ± 0.14	0.07 ± 0.01	43.15 ± 8.20	46.58	

Note: Concordant with what has been observed in the natural environment, the addition of NFRW from a natural spring to the seawater collected from the Malilnep and Guiguiwanen Channels in Bolinao, Pangasinan led to a decrease in salinity and an increase in inorganic nutrient concentrations.

water samples ($n = 4$ for each nutrient) were taken from the six salinity-nutrient treatments for nutrient analyses (i.e., NH_4^+, NO_3^-, NO_2^-, and PO_4^{3-}) following the methods below. The initial nutrient concentrations of the different salinity-nutrient treatments are presented in Table 1. Salinity-nutrient levels with TIN < 3 μM were considered low nutrient (LN) while those with TIN ranging from 5 to 6 μM and from 11 to 15 μM were labeled as medium nutrient (MN) and high nutrient (HN), respectively. The six treatments were then designated as O-LN-34, O-MN-30, O-HN-27, N-MN-30, N-HN-27, and N-HN-24, corresponding to water source-relative nutrient concentration-salinity. Prepared SW media were placed in separate plastic containers and stored in cold, dark conditions until use.

Source and acclimatization of the eucheumatoids

The biomass of *Kappaphycus alvarezii* (var. *tambalang*), *K. striatus* (var. *sacol-green*), and *Eucheuma denticulatum* (var. *spinosum*; Figure 2) used in this experiment were all sourced from the seaweed outdoor land-based nursery (OLBN) cultures of the Algal Ecophysiology Laboratory (AlgaE Lab) at the University of the Philippines—The Marine Science Institute-Bolinao Marine Laboratory (UPMSI-BML) in Bolinao, Pangasinan. The biomass of these three eucheumatoids was maintained in a glass aquaria supplied with air and nutrient-replete seawater from the Guiguiwanen Channel. All the eucheumatoids cultivated in the OLBN were from branch cuttings deposited at the Seaweed Culture Laboratory and Gene Bank (SCL&GB) of UPMSI and were maintained following the protocols described by Gonzaga et al. (2025). The "*tambalang*" variety of *K. alvarezii* and the "*sacol-green*" variety of *K. striatus* were originally sourced from Calatagan, Batangas in 2002 and 2021, respectively. The "*spinosum*" variety of *E. denticulatum* was originally collected from Tawi-Tawi in 2011. All species were molecularly confirmed following the barcoding described in Roleda et al. (2021).

For each species, the meristematic tips without visible epiphytes were excised (Figure 2 insets) and wounds were allowed to heal overnight in natural seawater inside a temperature-controlled room (25°C). For each experimental unit, ~2 g of starting biomass was used for each species.

Experimental treatment and setup

A total of 72 experimental units were prepared to accommodate the six treatment levels: O-LN-34, O-MN-30, O-HN-27, N-MN-30, N-HN-27, and N-HN-24 for the three species ($n = 4$ per treatment; 24 flasks per species). The acid-washed (10% HCl) 125-mL glass Erlenmeyer flasks were filled with 100 mL seawater culture media corresponding to the treatment level. The initial biomass used for each species was weighed to the nearest 0.0001 g using an analytical balance (Adventurer AR2140, Ohaus, United States). Thereafter, the ~2 g of acclimated *Kappaphycus alvarezii*, *K. striatus*, and *Eucheuma denticulatum* thalli were carefully placed into each flask. The flasks were randomly distributed and secured in three orbital shakers, each set to a speed of 200 revolutions per minute (rpm) to allow water motion and reduce the diffusion boundary layer (DBL), thus allowing efficient uptake of nutrients. Seawater media was changed every 24 h to avoid nutrient depletion and/or starvation. The salinity-nutrient experiment was conducted under controlled laboratory conditions at 29 ± 0.25°C temperature with a

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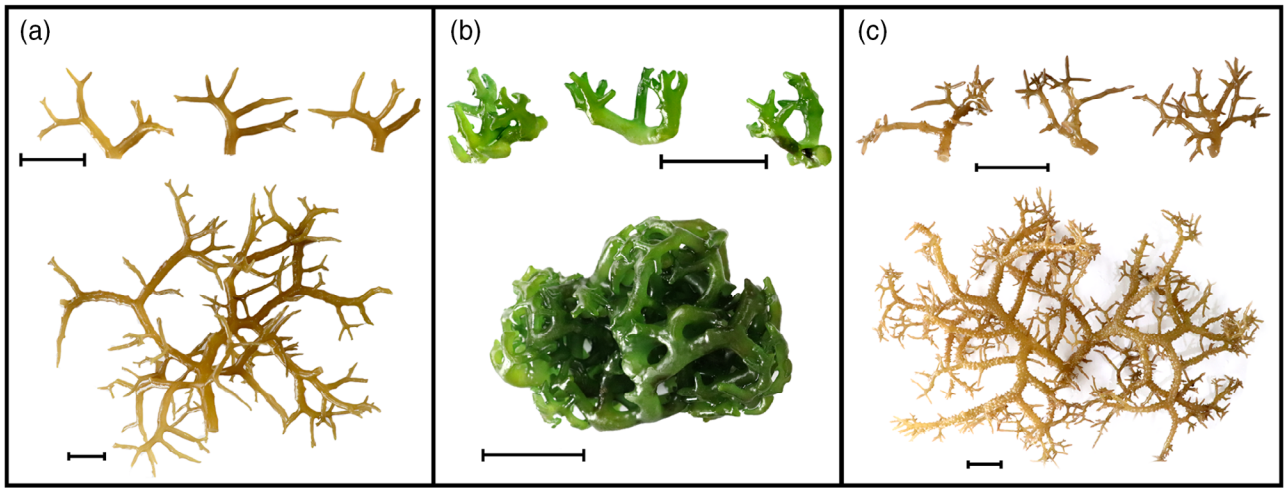


FIGURE 2 Young tips (top) excised from representative clonal plants (below) of the three eucheumatoid species cultivated in the OLBN used in the experiments. (a) *Kappaphycus alvarezii* (tambalang), (b) *K. striatus* (sacol-green), and (c) *Eucheuma denticulatum* (spinosum). Scale bars = 3 cm.

light intensity of 198 ± 12 μmol photons · m^{-2} · s^{-1} (measured using LI-14000, LiCOR, Nebraska, United States) emitted by LED lamps (LT8S-20 W-DL, Omni, Manila, Philippines) and a 12:12 h light:dark photoperiod. The experimental duration was 10 days, which was sufficient time to determine physiological and metabolic responses to environmental stressors.

Nutrient analyses and nutrient uptake

Nutrient uptake, measured as nutrient loss in the medium, was measured for each species after 24 h incubation corresponding to day 1 of the 10-day experimental cultivation period. After 24 h, seawater from each experimental unit was collected using a 50-mL syringe, and filtered through a 0.45-μm nylon syringe filter, filling 10 mL into each 15-mL conical tube for corresponding nutrient analysis (for 4 nutrients i.e., NH_4^+, NO_3^-, NO_2^-, and PO_4^{3-}) for analysis. All water samples were stored in a –20°C freezer until analysis. Nutrient analyses (see below) followed the standard protocols of Strickland and Parsons (1972) and Jones (1984).

Concentrations of NH_4^+, NO_3^-, NO_2^-, and PO_4^{3-} were all analyzed colorimetrically using a UV–Vis spectrophotometer (UV-1280, Shimadzu, Japan). Prior to the analysis, the frozen samples were thawed at room temperature. The NH_4^+ concentration of water samples was determined by first mixing 2.0 mL of trisodium citrate solution, 0.4 mL of phenol reagent, and 0.4 mL NaOCl reagent with the 10-mL water samples. The solution was then placed in a preheated oven at 70°C for 30 min. After the heating process, the water samples were allowed to cool down for 10 min before measuring the absorbance at 630 nm. The NH_4^+ concentration was calculated using the regression line (initial: $y = 0.0182x + 0.0159$, $r^2 = 0.9294$; final: $y = 0.0140x + 0.0047$, $r^2 = 0.9777$) obtained by plotting the absorbance of known concentrations of NH_4^+ (0–10 μM), which was prepared by dissolving an appropriate amount of NH_4Cl.

Nitrite concentration was analyzed by adding 0.2 mL of sulfanilamide solution to the 10-mL water sample and was mixed through manual shaking. After 2–8 min, a 0.2 mL N-1 solution was then added. The solution was allowed to react for 15 min until color development was observed. The absorbance was then measured at 540 nm. The NO_2^- concentration was calculated using the regression line (initial: $y = 0.0483x - 0.0006$, $r^2 = 0.9998$; final: $y = 0.0465x + 0.0008$, $r^2 = 0.9995$) generated by plotting the absorbance of known concentrations of NO_2^- (0–10 μM), prepared by dissolving an appropriate amount of $NaNO_2$.

The NO_x (NO_3^- + NO_2^-) concentration was analyzed by adding 2.0 mL of ammonium chloride buffer solution and 0.4 g of regenerated cadmium (Cd) pellets to the 10-mL water sample. The mixture was then placed in an orbital shaker at 200 rpm for 2 h. After shaking, 10 mL of the sample was decanted into new 15-mL conical tubes and augmented with 0.2 mL sulfanilamide solution. After 2–8 min, a 0.2 mL N-1 solution was then added. The solution was allowed to react for 15 min until color development was observed. The absorbance was then measured at 540 nm. The NO_x concentration was calculated using the regression line (initial and final: $y = 0.0287x - 0.0072$, $r^2 = 0.9946$) derived by plotting the absorbance of known NO_3^- concentrations (0–10 μM), prepared by dissolving an appropriate amount of $NaNO_3$. The NO_3^- concentration was calculated as the difference between NO_x and NO_2^- concentrations.

Phosphate concentration was determined by adding 1 mL of mixed reagent (i.e., ammonium molybdate, L-ascorbic acid, 3.2 M H_2SO_4, and potassium antimony tartrate) to the 10-mL water sample. The

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solution was left for 30 min until color development was observed. Absorbance was measured at 885 nm. The PO_4^{3-} concentration was calculated using the regression line (initial: $y = 0.0212x - 0.0010$, $r^2 = 0.9999$; final: $y = 0.0205x + 0.0022$, $r^2 = 0.9986$) obtained by plotting the absorbance of known PO_4^{3-} concentrations (0–10 μM), prepared by dissolving an appropriate amount of KH_2PO_4.

The uptake was expressed as a percentage of the amount of nutrients absorbed by the seaweed and was computed using the formula:

$$\%\,\text{Uptake} = \frac{N_i - N_f}{N_i} \times 100\%$$

where, N_i is the initial nutrient concentration, N_f is the final nutrient concentration.

Growth rate

Growth rates were measured after 3 (D3) and 10 (D10) days. Blotted-dried wet weights were weighed, and the percent daily growth rate (DGR = % · day^{-1}) was calculated using the recommended formula of Yong et al. (2013):

$$\text{DGR} = \left[\left(\frac{W_t}{W_o}\right)^{1/t} - 1\right] \times 100\%$$

where, W_t is the final weight in (g), W_o is the initial weight (g), and t is the culture duration (days).

Biochemical analyses

At the end of the 10-day experimental duration, samples from all experimental units were collected for biochemical analysis, that is, total polyphenol content, total soluble carbohydrate, and total protein. The samples were kept frozen in liquid nitrogen until processing. The total polyphenol content (TPC), total soluble carbohydrate, and total protein were measured colorimetrically using the standard Folin–Ciocalteu assay (Malick & Singh, 1980), anthrone assay, and Bradford assay (Bradford, 1976), respectively, following the protocols employed by Narvarte, Hinaloc, et al. (2022) but with some modifications (see below). Frozen samples were thawed and carefully pulverized in a clean mortar by using a pestle and liquid nitrogen. For each replicate, pulverized samples were then distributed to three conical tubes, corresponding to the three biochemical parameters to be measured. All colorimetrical assays were conducted using a UV–Vis spectrophotometer (UV-1280, Shimadzu, Japan).

Total polyphenol was extracted from ~0.1 g pulverized biomass samples using 10 mL of 80% methanol for 2 h in an orbital shaker set at 200 rpm (KJ201BD, Wincom Company Ltd., Changsha Hunan, China). Samples were then centrifuged for 20 min at 4000 rpm after the extraction process. After centrifugation, a 3mL aliquot of each extract was treated with 0.5 mL Folin–Ciocalteu reagent (which was made by 10× dilution of the concentrated reagent) and 2 mL of 8% Na_2CO_3 solution. Due to the formation of a precipitate, the solution was then centrifuged at 4000 rpm for 3 min, after which the bluish supernatant was carefully transferred to a cuvette for absorbance reading at 760 nm. The TPC was quantified using the regression line $y = 0.0424x + 0.0604$, $r^2 = 0.9737$, and was expressed as mg gallic acid equivalents per g fresh weight (mg GAE · g^{-1} FW).

Total soluble carbohydrate was extracted from ~0.1 g pulverized samples using 10 mL of distilled water placed in a preheated (80°C) water bath for 2 h. After the extraction, a 1 mL aliquot of the resultant solution was mixed with 4 mL of ice-cold anthrone solution (prepared by dissolving 200 mg of anthrone in 100 mL of sulfuric acid). The absorbance of the bluish-green complex was measured at 620 nm. Total soluble carbohydrate content was calculated using the regression line $y = 0.0069x - 0.0015$, $r^2 = 0.9999$ obtained using D-(+)-glucose as the calibration standard and was expressed as a percentage of the fresh weight (%FW).

Total protein was extracted from ~0.5 g powdered samples using 5 mL phosphate buffer containing 5 mM ethylenediamine tetraacetic acid (EDTA), 1 mM dithiothreitol (DDT), and 0.4% polypyrollidone. The solution was sonicated (UC-30A, Biobase) for 20 min and then centrifuged at 4000 rpm at room temperature. A 0.5 mL aliquot from the crude extract was reacted with 3 mL of Bradford reagent, resulting in a bluish complex. Absorbance was measured at 595 nm. Total protein content was computed using the regression line $y = 0.6128x - 0.0133$, $r^2 = 0.9990$ generated using bovine serum albumin (BSA) as standard and was expressed as the percentage of the fresh weight (%FW).

Data analyses

All data were presented as mean ± standard error (*SE*). Data on nutrient uptake, growth, and biochemistry were tested for normality using the Shapiro–Wilk test, and homogeneity of variance using Levene's test. Data were considered normal, and homogenous when $p > 0.05$. When data satisfied the assumptions of normality and homogeneity, their mean differences were tested using the parametric one-way analysis of variance (one-way ANOVA). Where significant differences ($p < 0.05$) were found, Tukey honestly significant difference (HSD) test was used as a post hoc test. For data that satisfied the assumptions of normality but failed to conform with the assumptions of homogeneity, their mean differences

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were tested using Welch one-way ANOVA, with Games–Howell test as a post hoc test. Data that did not satisfy the assumptions of normality were log-transformed. When data still did not conform to the assumptions of normality despite transformation, their mean differences were tested using the nonparametric Kruskal–Wallis test ($p < 0.05$). Where significant differences were found, a Dunn's test was employed as a post hoc test with Benjamini–Hochberg (BH) as a method for multiple pairwise comparisons. For the growth rate, two-way ANOVA was also employed to test which species showed the highest growth rate regardless of the treatments. All statistical analyses were conducted using R software version 4.3.1 (R Core Team, 2024).

RESULTS

Nutrient uptake

Nutrient uptake varied among the carrageenophytes and the inorganic nutrient species (Figure 3). Of the three inorganic nutrients (i.e., PO_4^{3-}, NH_4^+, NO_3^-), the uptake of NH_4^+ was the highest in all the eucheumatoids regardless of treatments, which ranged from 86% to 100% uptake.

In *Kappaphycus alvarezii*, PO_4^{3-} uptake ranging from 51% to 85% was not significantly different in all salinity-nutrient levels (Figure 3a; one-way ANOVA, $F_{(5,17)} = 1.508$, $p = 0.239$). For NH_4^+ (Kruskal–Wallis, χ^2 (5) = 11.597, $p = 0.0407$), the highest uptake (100%) was measured at high salinities regardless of nutrient levels, except at O-MN-27, and lower uptake (93%) at the lowest salinity with high nutrient (N-HN-24), but unfortunately there was no low-nutrient counterpart at this salinity level. For NO_3^- (one-way ANOVA, $F_{(3,12)} = 13.41$, $p = 0.0004$), the highest uptake (95%) was measured at low salinity and high nutrient (O-HN-27 and N-HN-24) and was lower (78%–93%) in all other treatment levels except that an increase in nutrient concentration in the media to 29% and 60% from the initial concentration was unexpectedly observed in two treatments (O-LN-34 and N-MN-30, respectively) levels.

In *Kappaphycus striatus*, uptake (range: PO_4^{3-} = 72%–89%; NH_4^+ = 87%–100%; NO_3^- = 61%–91%) in all nutrients monitored was not significantly different among the different salinity-nutrient levels (Figure 3b; PO_4^{3-} = Kruskal–Wallis, χ^2 (5) = 3.5099, $p = 0.6219$; NH_4^+ = Kruskal–Wallis, χ^2 (5) = 6.8736, $p = 0.2302$; NO_3^- = Kruskal–Wallis, χ^2 (3) = 7.3897, $p = 0.0605$). The same increase in NO_3^- concentration in the media was observed at O-LN-34 and N-MN-30 (32% and 52%, respectively). Conversely, in *Eucheuma denticulatum*, only uptake in PO_4^{3-} (49%–87%) and NH_4^+ (86%–100%) were not significantly different among the different salinity-nutrient levels (Figure 3c; PO_4^{3-} = Kruskal–Wallis, χ^2 (5) = 6.384, $p = 0.2706$; NH_4^+ = Kruskal–Wallis, χ^2 (5) = 7.9779, $p = 0.1575$). For NO_3^- (Kruskal-Wallis, χ^2 (5) = 17.354, $p = 0.0039$), uptake was highest at O-HN-27 (97%). At the lower salinity of 27, percent uptake was comparable between the high (N-HN-27 = 95% and O-HN-27 = 97%) nutrient levels regardless of the seawater source and also at much lower salinity but high nutrient concentration (N-HN-24 = 93%). Nitrate uptake was lowest (41%) at the ambient salinity of 30 with medium nutrient concentration but was comparable with the ambient salinity of 34 with low nutrient concentration.

Growth

After 3 days (Figure 4a,c,e), regardless of salinity-nutrient treatment, growth was highest in *Kappaphycus alvarezii* (3.34%–6.33%), followed by *K. striatus* (2.13%–3.54%), and lowest in *Eucheuma denticulatum* (1.28%–2.79%; two-way ANOVA, $F_{(2,64)} = 75.57$, $p < 0.001$). At day 10 (Figure 4b,d,f) growth rates of both *Kappaphycus* species were comparable (*K. alvarezii* = 1.99%–3.36% and *K. striatus* = 2.35%–3.16%) and still lowest in *E. denticulatum* (1.31%–2.20%; two-way ANOVA, $F_{(2,64)} = 45.50$, $p \leq 0.001$).

At different salinity-nutrient levels during D3, the growth rate of *Kappaphycus alvarezii* (Figure 4a; one-way ANOVA, $F_{(5,18)} = 10.78$, $p \leq 0.001$) was highest at N-HN-24 and lowest at O-LN-34. At salinity 27, varying nutrient concentrations did not elicit a significant effect. However, at D10 (Figure 4b; one-way ANOVA, $F_{(5,18)} = 14.34$, $p \leq 0.0001$), the lower salinities with high nutrient concentrations (N-HN-24 and N-HN-27) measured the highest growth rates compared with higher salinities with low and medium nutrient concentrations (O-LN-34 and O-MN-30) as well as the high nutrient salinity 27 with ambient seawater collected from the offshore (O-HN-27; Figure 4b).

In *Kappaphycus striatus* at D3 (Figure 4c; Welch one-way ANOVA, $F_{(5,18)} = 9.1453$, $p = 0.0048$), the growth rate at the lowest salinity with high nutrient concentrations (N-HN-24) was only significantly higher than O-HN-27 and not the other treatments. However, at D10 (Figure 4d; Welch one-way ANOVA, $F_{(5,18)} = 3.0272$, $p = 0.0861$), the growth rates did not differ significantly among treatments.

Interestingly, in *Eucheuma denticulatum* at D3 (Figure 4e; Welch one-way ANOVA, $F_{(5,18)} = 32.272$, $p = < 0.0001$), the growth rates were similar among lower salinity but high nutrient treatments (N-HN-24, N-HN-27, and O-HN-27) regardless of the ambient seawater source. Growth rates were lowest at higher salinities regardless of nutrient concentrations (O-LN-34, O-MN-30, and N-MN-30). At D10 (Figure 4f; Kruskal–Wallis, χ^2 (5) = 19.27, $p = 0.0017$), all treatments with medium and high nutrient concentration (O-MN-30, O-HN-27, N-HN-27, and N-HN-24) showed comparable growth rates except N-MN-30.

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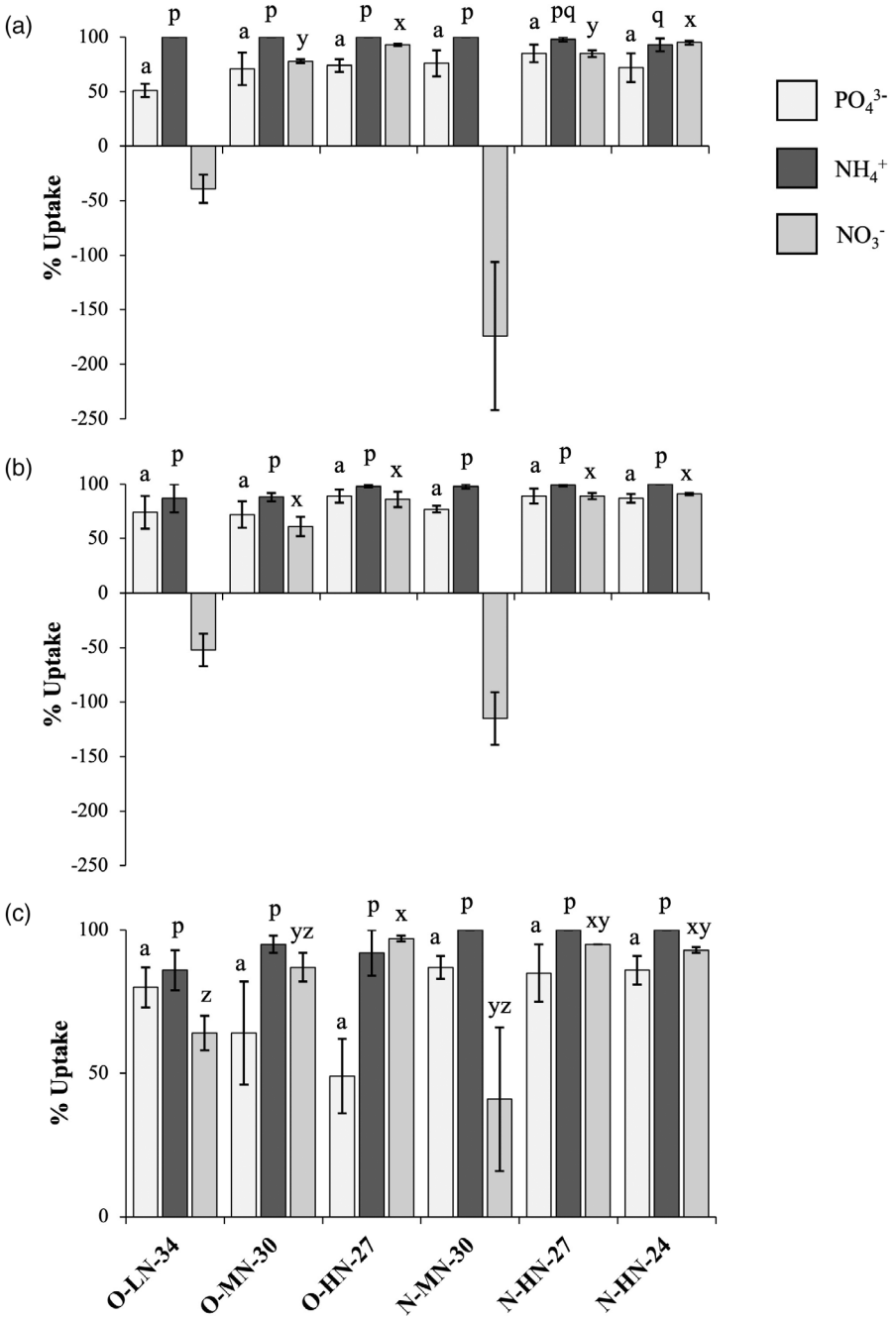


FIGURE 3 Nutrient uptake (mean ± *SE*) capacity of three eucheumatoids, expressed as % uptake, under the different salinity-nutrient treatments during the day 1. (a) *Kappaphycus alvarezii*, (b) *K. striatus*, and (c) *Eucheuma denticulatum*. Negative nitrate uptake at O-LN-34 and N-MN-30 for both *Kappaphycus* species due to increase in the final concentration of the water media potentially due to "leaking." Different letters in each species represent significant mean differences ($p < 0.05$). Three letters are assigned for each nutrient, that is, a, b, and c (PO_4^{3-}); p, q, and r (NH_4^+), and x, y and z (NO_3^-). *x*-axis labels correspond to treatments indicating seawater source (O: Offshore; N: Nearshore), relative nutrient concentrations (LN: Low nutrient; MN: Medium nutrient; HN: High nutrient) and corresponding salinity (34, 30, 27, and 24).

Biochemistry

The means of the total polyphenol content, total soluble carbohydrate, and total protein in *Kappaphycus alvarezii* and *K. striatus* were not significantly different across all treatments (Table 2; KA: TPC, one-way ANOVA, $F_{(5,15)} = 0.912$, $p = 0.499$; TSC, one-way ANOVA, $F_{(5,17)} = 0.931$, $p = 0.486$; TP, one-way ANOVA, $F_{(5,17)} = 2.175$, $p = 0.105$; KS: TPC, one-way ANOVA, $F_{(5,17)} = 2.436$, $p = 0.0772$; TSC, Kruskal–Walis, χ^2 (5) = 3.305, $p = 0.653$; TP, one-way ANOVA, $F_{(5,18)} = 0.397$, $p = 0.844$). For *Eucheuma denticulatum*, mean total protein was also comparable across all treatments (one-way ANOVA, $F_{(5,18)} = 0.762$, $p = 0.589$). However, the mean total polyphenol content significantly differed across treatments (one-way ANOVA, $F_{(5,16)} = 4.174$, $p = 0.0128$) and was higher in the higher nutrient condition (HN) regardless of salinity (24–30) and lowest at the highest salinity with lower nutrient concentration (O-LN-34). The same trend was observed in the mean total carbohydrate (Kruskal–Wallis, χ^2 (5) = 11.006, $p = 0.0513$).

DISCUSSION

Freshwater outflows alter seawater salinity by creating a mesocosm or stratified hyposaline condition with concomitant higher inorganic nutrient levels in the

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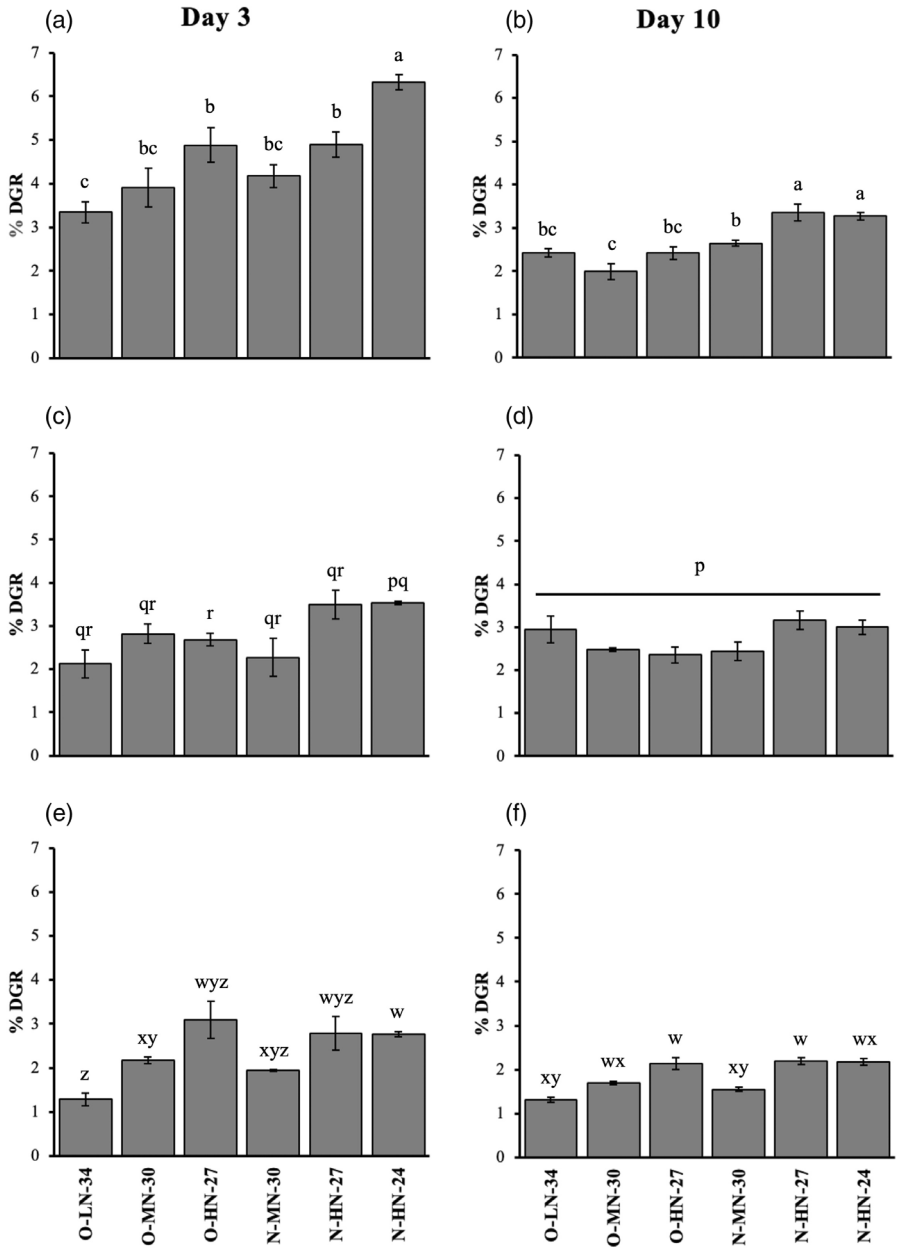


FIGURE 4 Growth rates (mean ± *SE*) of *Kappaphycus alvarezii* (a, b), *K. striatus* (c, d), *Eucheuma denticulatum* (e, f) at day 3 (a, c, e), and day 10 (b, d, f) under the different salinity-nutrient treatments. Different letters in each species per monitoring period represent significant mean differences ($p < 0.05$). *x*-axis labels correspond to treatments indicating seawater source (O: Offshore; N: Nearshore), relative nutrient concentrations (LN: Low nutrient; MN: Medium nutrient; HN: High nutrient), and corresponding salinity (34, 30, 27, and 24).

coastal marine environment. Our study demonstrated that high nutrient concentrations in low salinity media did not elicit the typical negative low salinity effects on seaweed physiology and chemistry. The perpetuation of physiological integrity in *Kappaphycus alvarezii*, *K. striatus*, and *Eucheuma denticulatum* provides evidence in support of the hypothesis proposed by Roleda and Hurd (2019) that the interactive effects of high nutrient and low salinity can compensate for the independent negative effects of low salinity. Nutrient uptake varied among inorganic nutrient species and different eucheumatoid species but showed a preference for NH_4^+ regardless of the treatment. Although NO_3^- uptake was generally higher in treatments with higher NO_3^- concentration, growth rates were species-specific but generally higher at lower salinities with higher nutrient concentrations. Total soluble carbohydrate, total protein, and total polyphenol content did not significantly vary among treatments, except in *E. denticulatum*. Our results provide new insights into the potential of farming eucheumatoids in nearshore areas with fluctuating salinity but with corresponding high nutrient levels as reported by different seaweed farmers.

Our LN treatment (Table 1) results suggest that the different macronutrient concentrations were still sufficient to support the daily nutrient requirements of the species, as the media were replenished every 24 hours. In a relatively pristine reef flat in front of Silaki Island in Bolinao, northwestern Philippines, concentrations of NO_x (NO_3^- + NO_2^-), NH_3, and PO_4^{3-} range between 0.01 and 0.39 μM, 1.7 and 5.1 μM, and 0.3 and 0.8 μM, respectively (Senal et al., 2011). In a eucheumatoid farm site along the coast of Bulusan, Sorsogon, Philippines,

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TABLE 2 Biochemistry of the three eucheumatoids after 10 days of cultivation in different salinity-nutrient treatments (mean ± *SE*).

Species	Salinity-nutrient treatments	Biochemical components		
		Total polyphenol content (mg GAE · g^{-1} FW)	Total soluble carbohydrate (%FW)	Total protein (%FW)
Kappaphycus alvarezii (“*tambalang*”)	O-LN-34	0.08 ± 0.02^{a}	2.99 ± 0.33^{a}	0.09 ± 0.01^{a}
	O-MN-30	0.23 ± 0.16^{a}	3.52 ± 0.84^{a}	0.10 ± 0.01^{a}
	O-HN-27	0.04 ± 0.03^{a}	4.96 ± 2.21^{a}	0.07 ± 0.00^{a}
	N-MN-30	0.10 ± 0.05^{a}	3.78 ± 0.94^{a}	0.09 ± 0.01^{a}
	N-HN-27	0.06 ± 0.03^{a}	2.18 ± 0.14^{a}	0.09 ± 0.01^{a}
	N-HN-24	0.10 ± 0.05^{a}	2.80 ± 0.12^{a}	0.08 ± 0.01^{a}
K. striatus (“*sacol-green*”)	O-LN-34	0.12 ± 0.01^{p}	3.49 ± 1.28^{p}	0.11 ± 0.01^{p}
	O-MN-30	0.43 ± 0.22^{p}	2.22 ± 0.33^{p}	0.13 ± 0.02^{p}
	O-HN-27	0.21 ± 0.05^{p}	2.99 ± 0.40^{p}	0.11 ± 0.02^{p}
	N-MN-30	0.35 ± 0.08^{p}	4.03 ± 1.19^{p}	0.10 ± 0.01^{p}
	N-HN-27	0.25 ± 0.06^{p}	4.72 ± 2.12^{p}	0.11 ± 0.03^{p}
	N-HN-24	0.39 ± 0.15^{p}	3.06 ± 0.31^{p}	0.10 ± 0.01^{p}
Eucheuma denticulatum (“*spinosum*”)	O-LN-34	0.25 ± 0.02^{x}	1.44 ± 0.74^{x}	0.20 ± 0.01^{x}
	O-MN-30	0.29 ± 0.04^{xy}	4.35 ± 0.62^{y}	0.18 ± 0.03^{x}
	O-HN-27	0.33 ± 0.03^{xy}	3.91 ± 0.30^{y}	0.20 ± 0.01^{x}
	N-MN-30	0.38 ± 0.01^{y}	3.45 ± 0.18^{xy}	0.18 ± 0.02^{x}
	N-HN-27	0.37 ± 0.02^{y}	4.08 ± 0.13^{y}	0.24 ± 0.05^{x}
	N-HN-24	0.38 ± 0.01^{y}	3.49 ± 0.22^{xy}	0.23 ± 0.04^{x}

Note: Different letters in each species represent significant mean differences ($p < 0.05$). Three letters are assigned for each eucheumatoid species, that is, a, b, and c (*K. alvarezii*); p, q, and r (*K. striatus*), and x, y and z (*E. denticulatum*).

macronutrients, NO_2^-, NH_4^+, and PO_4^{3-} range between 0.08 and 5.27 μM, 0.10 and 0.31 μM, and 0.35 and 0.83 μM, respectively (Villanueva et al., 2011). The above information suggests that eucheumatoid farming can thrive in coastal waters with sufficient background nutrient levels but with corresponding seasonal high and low biomass production relative to the ambient nutrient concentration. Likewise, with daily change in culture media in our experiment, the plants were relatively well nourished.

Species-specific nutrient uptake and preference for certain nitrogen species as observed in this study have been well reported. For example, a recent study on *Kappaphycus alvarezii*, *K. striatus*, and *Eucheuma denticulatum* showed a consistently higher uptake of NH_4^+ compared to NO_3^-, indicating a preference for NH_4^+ (Narvarte et al., 2025; Narvarte & Roleda, 2025). In the carrageenophyte *Hypnea musciformis*, NH_4^+ was rapidly taken up over NO_3^- (Haines & Wheeler, 1978). Similarly, the agarophyte *Gracilaria tenuistipitata* also demonstrated a preference for NH_4^+ by assimilating it more rapidly despite NO_3^- availability (Wang et al., 2014). The NH_4^+ preference of several macroalgal species is related to its passive diffusion uptake mechanism and the lesser energetic cost during assimilation (Harrison & Hurd, 2001; Roleda & Hurd, 2019; Ross et al., 2018).

It was noteworthy and unexpected to observe an increase in NO_3^- concentration in the media of both *Kappaphycus* species but not *Eucheuma denticulatum* after 24 h of incubation under the ambient salinities of 34 and 30 (without freshwater dilution) from Malilnep (offshore) and Guiguiwanen (nearshore) waters, respectively. Considering that a general trend was observed, these measurements are unlikely to be artifacts. One possible explanation for the negative % uptake rates (i.e., due to the increase in NO_3^- concentration in the media) in both *Kappaphycus* species is that they leaked dissolved organic matter (DOM) or particularly dissolved organic nitrogen (DON) under these ambient salinities, which were remineralized by microbes into inorganic nitrogen. However, the trigger for such a possible leakage phenomenon under the non-stressful conditions for both *Kappaphycus* species is unknown. A similar case of “leaking” nitrogen has been previously reported in *Ulva expansa*, but under insufficient nitrogen conditions, contributing to its eventual senescence by releasing its tissue nitrogen-pool into the environment for other species to utilize (Fong et al., 1996). Moreover, the release of NO_3^- into the bulkwater (“leaking”) has also been reported in *Porphyra leucosticta* (=*Pyropia leucosticta*) during the first few hours of exposure to low nutrient concentration media (3 μM NO_3^-) at 15°C, thereby resulting in a negative uptake rate, but NO_3^- uptake commenced after 3 h of incubation (Pedersen et al., 2004). Leakage of NO_3^- in both *Kappaphycus* species at O-LN-34 and N-MN-30 with initial bulkwater NO_3^- levels of <1 μM (Table 1), but not

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in *E. denticulatum*, is noteworthy. A previous study suggested that *E. denticulatum* can efficiently take up NO_3^- even at very low levels (Narvarte, Genovia, et al., 2022). Nonetheless, the most important results suggest that all eucheumatoids in this study are nitrogen-limited and have a preference for NH_4^+ uptake. Further studies are necessary to elucidate the factors and mechanisms of DOM and DON leakage and remineralization, as our observations are likely not artifacts.

Growth rates of the three eucheumatoids were generally higher at the lower salinity range of 27–24, provided that the plants were supplied with sufficient macronutrients: TIN ranging from 2 to 14.56 μM and phosphate from 0.6 to 1.89 μM (Table 1). Conversely, at a higher salinity range of 30–34, the lower concentration range of TIN at 2.4–6.3 μM despite the comparable PO_4^{3-} at 0.6–1.9 μM, growth rates were lower. This result suggests that nitrogen is the limiting nutrient for this species. Several seaweeds have an nitrogen:phosphorus ratio of 30:1, indicating that they require more nitrogen than phosphorus, making them prone to nitrogen limitation (Atkinson & Smith, 1983; Harrison & Hurd, 2001). Moreover, the combined effects of high nutrients and low salinity on growth have also been demonstrated in other macroalgal species. For example, the rhodophyte *Gracilaria salicornia*, growing in a volcanic island with SGD characterized by low salinity but nutrient-rich water and sufficient dissolved inorganic carbon (DIC), measured increased carbon uptake and growth (La Valle et al., 2023). Likewise, *G. tenuistipitata* bloomed when grown under low salinity (~12) in a simulated brackish environment with high nitrogen concentration (Wang et al., 2014). In the chlorophyte *Enteromorpha intestinalis* (=*Ulva intestinalis*), the highest biomass was recorded in nitrogen-enriched water, albeit in low salinity (15) conditions (Kamer & Fong, 2001). With sufficient nitrogen supply, the same species also showed a competitive advantage over *U. expansa* at low 15 salinity (Fong et al., 1996). However, their findings must be interpreted with caution since *U. intestinalis* can thrive in oligohaline (0.5–5) to euhaline (30–40) environments (Rybak, 2018), indicating its tolerance to a wide salinity range regardless of the nutrient levels. In contrast, wild populations of these eucheumatoids (e.g., *Kappaphycus alvarezii, K. striatus*, and *Eucheuma denticulatum*) are typically observed in euhaline environments. Nevertheless, exposure to low salinity with a corresponding increase in inorganic nutrients promotes the growth of the three eucheumatoids species examined in this study.

Under a salinity of 24, the growth rate (6.33%) of *Kappaphycus alvarezii* (this study) after 3 days of cultivation was 12-fold higher compared to the growth rate (>0.5%) reported by Siddiqui et al. (2024) after 5 days of cultivation. This huge discrepancy in the reported growth rates under low salinity may be attributable to the differences in the nutrient levels in the culture media. In the previous study, Milli-Q water, which is devoid of important minerals and nutrients, was used to prepare hyposaline culture media. This media preparation lacks ecological significance, as it not only reduces salinity levels but also dilutes the available nutrients and minerals atypical of salinity reduction due to rainfall or freshwater outflow. On the contrary, our study used natural spring water with naturally high nutrient concentrations to prepare hyposaline culture media that was ecologically relevant. Therefore, our study and results highlight the importance of preparing ecologically relevant hyposaline media to study the salinity stress physiology of algae.

Recently, wild *Kappaphycus* cf. *striatus* was reported to thrive in a mangrove area in the southern Philippines, exposed to fluctuating salinity characteristics of an estuarine habitat (Galera et al., 2023). It is very likely that the water there also contained high inorganic nutrients due to the mineralization of particulate organic matter from the deposited litterfall of mangrove vegetation, tidal import of inorganic nutrients, and inputs from terrestrial run-offs (Taillardat et al., 2019). Nevertheless, the result of this study has provided empirical evidence on the broader range of salinity tolerance of *K. striatus*, confirming their potential to be farmed in intertidal areas exposed to the elements or in areas with soft bottoms near mangrove forests as observed in farms in northern Bohol. Meanwhile, there are limited studies on the salinity tolerance of *Eucheuma denticulatum*. According to Largo et al. (1995), a broad salinity range from 25 to 35 is favorable for this species, while 20 or lower has been observed to be fatal. The high growth rates of this species at lower salinities (24–27) with sufficient nutrient concentration in this study partially support the premise of the previous report but contradict the statement on the positive effect of higher salinity (e.g., 30–34), which was not evident in this study. For future studies, it is recommended to test the response of *E. denticulatum* at much lower salinities (e.g., <20) under nutrient-replete and deplete conditions.

A taxa-specific response was observed in the biochemical profiles of the three species investigated. Although the polyphenol, carbohydrate, and protein contents of both *Kappaphycus* species did not vary across all treatment levels, the mean total polyphenols and total carbohydrate contents of *Eucheuma denticulatum*, although not pronounced, exhibited indicative inverse relationships with nutrient concentration. Considering that all the HN, MN, and LN conditions contained relatively sufficient TIN and phosphate, that is, were replenished daily, and that seawaters from Malilnep and Guiguiwanen and freshwater *subor* spring are known to be DIC saturated (Isah et al., 2022), our experimental cultivation was literally conducted under sufficient macronutrients (carbon, nitrogen, and phosphorus) conditions. This explains the no significant variations in the mean total protein contents in all species investigated and in the total polyphenol and carbohydrate in both *Kappaphycus* species. Although tissue protein and

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carbohydrate contents may primarily be influenced by the relative concentrations of the macronutrients carbon and nitrogen, respectively, there is an indication that total phenolic concentration (TPC) of free phenolic compounds in land plants (i.e., wheat) increases with increasing nitrogen supply; however, the TPC of conjugated soluble phenolics has decreased with increasing nitrogen (Stumpf et al., 2015). The application of nitrogenous, phosphorus, and potassium fertilizers, likewise, significantly increased the total polyphenols in bush tea (Mudau et al., 2007). In marine plants, ammonium enrichment ($15\,\mu mol \cdot L^{-1}$) of the seagrass *Thalassia hemprichii* significantly increased its total phenolic content but decreased with further enrichment to 45 and $60\,\mu mol \cdot L^{-1}$ (Fang et al., 2022). In seaweeds, the role of macronutrients, for example, nitrogen and ph, in the biosynthesis of polyphenols remains to be elucidated. Our data, however, suggest that considering the higher and efficient NO_3^- uptake in *E. denticulatum* compared to *Kappaphycus* spp. (Narvarte, Genovia, et al., 2022), the observed positive correlation between polyphenol content and nutrient concentration is supported.

In summary, the growth of the three eucheumatoid species at a lower salinity of 24 but with a high nutrient concentration was better compared with the ambient seawater salinities (i.e., 30 and 34). This concurs with the observations and anecdotal reports of seaweed farmers in northern Bohol, Philippines, which has important implications for the future of eucheumatoid seaweed farming. Moreover, this study provides fresh insights into the potential of expanding seaweed farms beyond the traditionally recommended sites using resilient species and/or the cultivars used in our study. Although the results of this study look promising in the context of seaweed farming, field trials are still necessary, considering that a confluence of other environmental factors (e.g., light and temperature) will be at play, which could confirm whether these eucheumatoids will grow and thrive in nearshore areas with fluctuating salinity caused by the inflow of nutrient-rich freshwater.

AUTHOR CONTRIBUTIONS

Jonh Rey L. Gacura: Conceptualization (supporting); data curation (lead); formal analysis (lead); investigation (lead); methodology (equal); validation (lead); visualization (lead); writing – original draft (lead); writing – review and editing (lead). **Bienson Ceasar V. Narvarte:** Conceptualization (supporting); data curation (supporting); formal analysis (supporting); investigation (lead); methodology (equal); validation (supporting); writing – review and editing (equal). **Emmanuel M. Mendoza:** Data curation (supporting); investigation (lead); methodology (supporting); validation (supporting); writing – review and editing (supporting). **Lourie Ann R. Hinaloc:** Data curation (supporting); investigation (supporting); methodology (equal); validation (supporting); writing – review and editing (equal). **Ronel T. Aguilar:** Investigation (supporting); methodology (supporting); validation (supporting); writing – review and editing (supporting). **Eugene Yolanda Irang-Maxion:** Investigation (supporting); validation (supporting); writing – review and editing (supporting). **Shienna Mae C. Gonzaga:** Data curation (supporting); investigation (supporting); validation (supporting); writing – review and editing (supporting). **Michael Y. Roleda:** Conceptualization (lead); funding acquisition (lead); investigation (supporting); methodology (lead); project administration (lead); resources (supporting); supervision (lead); validation (supporting); writing – review and editing (lead).

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DATA AVAILABILITY STATEMENT

Data will be made available upon request from the authors.

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